Interpretable Sentinel-2 Turbidity Mapping with Ensemble Symbolic Regression

A Reproducible Visual-Programming Pipeline for the Lower Seine

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Abstract

We present a **proof-of-concept** pipeline for estimating water turbidity (NTU) from Sentinel-2 surface reflectance using interpretable models and an explicit ensemble uncertainty. The pipeline is built as a visual node graph inside the open-source environment **VNStudio**, exposing every parameter on a connectable node and serializing the entire experiment to a single JSON file. Training uses 563 satellite–station matchups assembled from the French Hub'Eau Naïades network for the lower Seine basin (2017–2025). We benchmark six model families (genetic-programming symbolic regression with and without rebalancing or sample weighting, RandomForest, HistGradientBoosting) under a unified 10-fold cross-validation. The GP family fails to generalise (out-of-fold $R^2 \leq 0$ across every configuration); tree ensembles reach $R^2 \approx 0.22$ with a predicted-vs-observed slope of ≈ 0.24. The best in-fold GP expression nonetheless rediscovers the Blue/Red ratio and NIR/SWIR contrast that anchor the published bio-optical models, supporting their physical relevance. Applied to a 10 m July 2023 acquisition over the Rouen reach, the RandomForest field returns a median turbidity of 14 NTU with a per-pixel tree-spread σ providing operational uncertainty. We position the work as (i) a methodological demonstration of *visual programming as a reproducibility primitive*, (ii) an honest diagnosis of class-imbalance brittleness in symbolic regression on long-tailed environmental targets, and (iii) a starting point for the improvements outlined in our future-work section.

Keywords: turbidity, Sentinel-2, symbolic regression, gplearn, ensemble uncertainty, Naïades, Seine, visual programming.

1 Introduction

Operational satellite mapping of inland-water turbidity relies overwhelmingly on either fixed empirical formulas [4, 2] or end-to-end machine-learning regressors [5, 6]. The former are interpretable but inflexible; the latter are flexible but opaque. Genetic-programming symbolic regression [3, 7] sits between the two: it searches a space of closed-form expressions, returning a model that a hydrologist can inspect, criticize, and connect to physical reasoning.

We pair GP with two practical ingredients that the literature usually treats separately. First, an *ensemble* of independently seeded runs whose spread quantifies model uncertainty per pixel. Second, a *visual-programming* substrate (VNStudio) in which the entire workflow — API call, atmospheric correction, sampling, feature engineering, training, inference, raster reconstruction, statistics — is a single node graph. The visual layer is not cosmetic: it locks each preprocessing

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decision to a parameter on a node, and exports the whole experiment as a JSON file that can be re-opened and re-executed.

Our contribution is four-fold:

1. A reproducible Sentinel-2 turbidity pipeline whose entire state (parameters, connections, in-situ source) lives in a single 14-node graph (Fig. 1).
2. Ensemble symbolic regression with explicit per-pixel uncertainty, applied to the lower Seine over July 2023.
3. Empirical evidence that an unconstrained GP search rediscovers the Blue/Red and NIR/SWIR structural building blocks of established turbidity formulas, supporting their physical relevance.
4. A concrete diagnosis of *class-imbalance collapse* in symbolic regression on long-tailed environmental targets, with a stratified-subsampling remedy implemented inline in the graph.

This work is positioned as a **proof-of-concept**: the model's quantitative skill remains modest (slope 0.26, predictions saturating well below the upper Naïades range), and we explicitly do not claim operational readiness. Rather, the contribution is to expose a reproducible substrate on which improvements — sample weighting, larger geographic transfer, calibrated uncertainty — can be tested in isolation without rewriting the rest of the pipeline (§7).

2 Study area and data

2.1 Geographic scope

We focus on the lower Seine basin from the Île-de-France down to the estuary near Rouen. The *training* bounding box is wide — `[-0.0225, 48.5583, 2.8878, 49.6460]` (WGS84, lon/lat) — to maximize Naïades station coverage. The *inference* tile, chosen for visual detail, is a 34 km-wide window over Rouen: `[0.9522, 49.2499, 1.2621, 49.4734]`, resolved at 10 m.

2.2 Satellite data

Sentinel-2 L2A scenes were retrieved through the Copernicus Data Space catalogue (`geo_copernicus` node). Acquisition window: 2023-07-01 to 2023-07-07; the lowest-cloud scene (2023-07-04) was selected automatically. Six surface-reflectance bands were extracted: B02 (Blue, 492 nm), B03 (Green, 560 nm), B04 (Red, 665 nm), B08 (NIR, 833 nm), B11 (SWIR1, 1614 nm) and B12 (SWIR2, 2202 nm). Maximum scene-level cloud cover was capped at 10 %.

2.3 In-situ measurements

Turbidity ground truth comes from the French Hub'Eau Naïades API (`/api/v2/qualite_rivieres/analyse_p` parameter code `1295`, Formazine NTU). To overcome the per-call ceiling of 20 000 rows, the `geo_naiades` node chunks queries by year over the 2017–2025 window. Only validated measurements (`code_remarque=1`) are retained. The raw download returned 563 station-day matchups falling within ±30 d of the satellite acquisition.

3 Methods

3.1 Pipeline overview

The pipeline (Fig. 1) splits into a *training* branch (top) and an *inference* branch (bottom), sharing a single GP ensemble. Sections 3.2–3.9 document each component and its parameters.

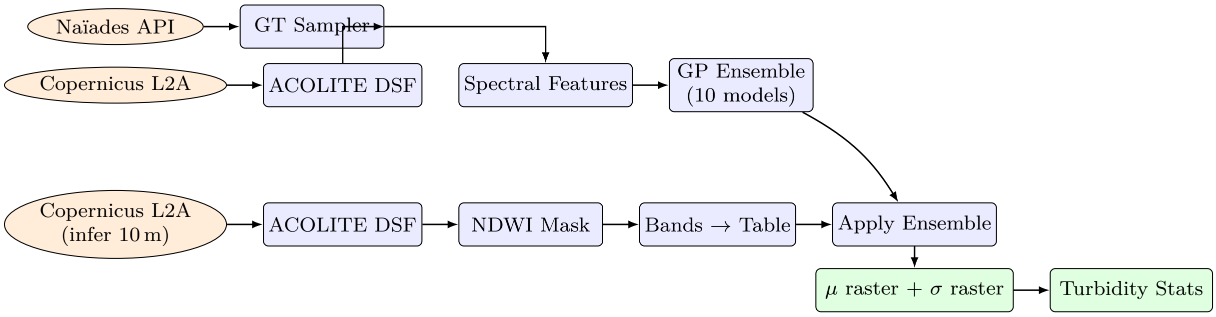


Figure 1: Pipeline graph as instantiated in VNStudio (schematic). Each box is a node; each arrow a typed connection (geotiff, dataframe, mask). The full graph is shipped as `public/templates/Machine Learning Course/gp_ensemble_uncertainty.vn`.

3.2 Atmospheric correction (`geo_acolite_full`, DOS-1 mode)

Raw L2A reflectance still carries residual aerosol scattering and surface effects. We apply a fast Dark-Object Subtraction (DOS-1) variant on the L2A stream, configured with `band_names = Blue,Green,Red,NIR,SWIR1,SWIR2` and `dsf_aot_estimate = dark_spectrum`. Two implementation choices are non-obvious:

- **Auto-scaling**: the test for "raw DN vs. reflectance" uses the 99th-percentile rather than the maximum, which would be dragged by a single saturated pixel and divide the entire scene by 10 000.
- **Dark-object selection**: the dark value is the 0.1^{th} percentile of positive pixels, capped to discard cases where the 1^{st} percentile coincides with the water signal. When dark $>$ $0.5 \times$ median the subtraction is skipped.

Output is Rrs $\in [0, 0.2]$ sr^{-1}, directly comparable to in-situ Rrs.

3.3 Ground-truth sampling (`geo_ground_truth_sampler`)

The Naïades DataFrame is joined to the satellite raster through lat/lon $\rightarrow$ pixel mapping. Each station is sampled via a 5×5 `nanmean` window (smoothing sensor noise and co-registration error) and constrained by a ± 30 d temporal window around the satellite date. Parameters:

Parameter	Value
`lat_col / lon_col`	`lat / lon`
`target_col`	`label` (NTU)
`band_names`	Blue, Green, Red, NIR, SWIR1, SWIR2
`target_min / target_max`	0 / 500 NTU
`mask_radius`	3 px
`sample_window`	5 px (5×5 nanmean)
`image_date`	2023-07-04
`date_window`	30 d

3.4 Spectral feature engineering (`ml_spectral_features`)

Identical on the training and inference branches:

- Ratios: `Red/Green`, `Blue/Green`, `Red/NIR`, `NIR/SWIR1`, `Red/SWIR1`
- Indices: NDTI $= (R - G)/(R + G)$, MNDWI $= (G - \text{SWIR1})/(G + \text{SWIR1})$
- Log transforms: $\log(\text{Red}), \log(\text{SWIR1})$

This expanded basis enlarges the symbolic search space without forcing a particular model shape.

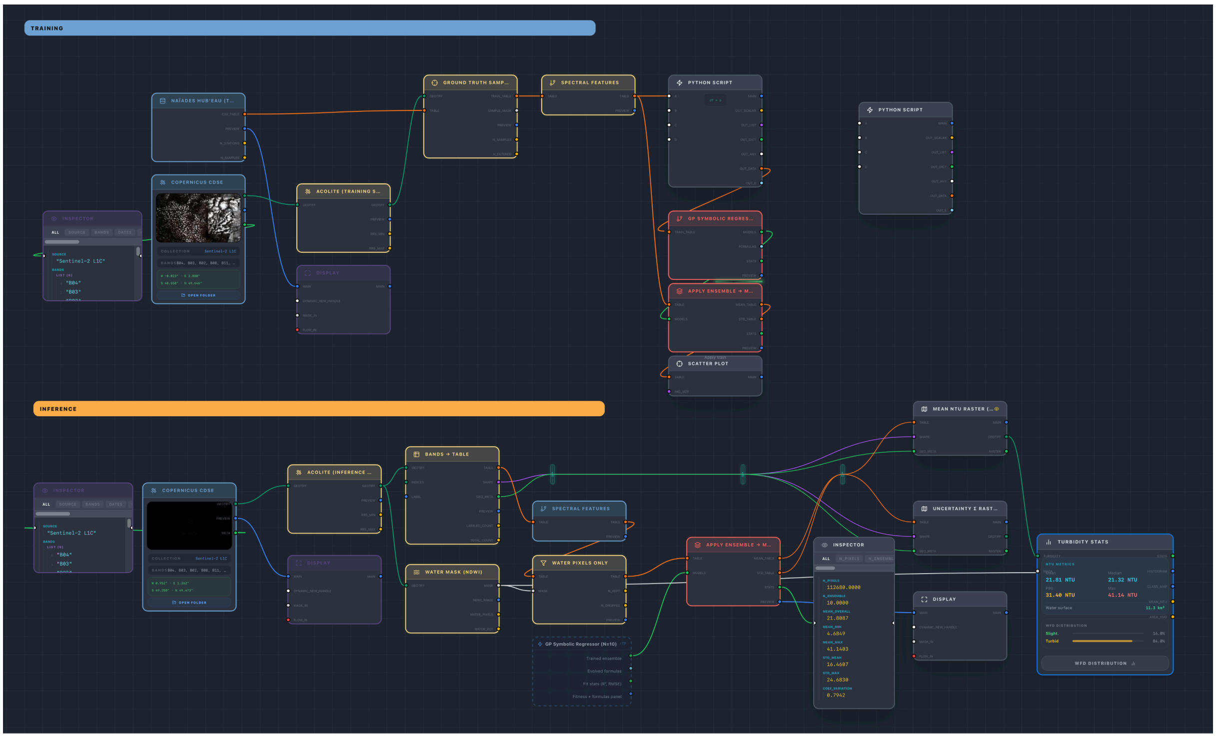


Figure 2: Actual VNStudio canvas screenshot. Top swimlane: training branch (Naïades download → ACOLITE → Ground-Truth Sampler → Spectral Features → logic_python rebalancer → GP Ensemble). Bottom swimlane: inference branch (Copernicus → ACOLITE → NDWI mask → Bands-to-Table → Apply Ensemble → Mean/Std rasters → Turbidity Stats). The entire experiment state lives in the JSON serialization of this canvas.

3.5 Symbolic regression ensemble (`ml_symbolic_regressor`)

We run `gplearn` 10 times with seeds $\{423, 424, \ldots, 432\}$. Common parameters:

Parameter	Value
`target_column`	`label`
`n_ensemble`	10
`population_size`	1000
`generations`	50
`parsimony`	0.001
`test_size`	0.20
`log_transform`	True
`use_log / use_sqrt / use_div`	True / True / True

The function set is $\{+, -, \times, \div_{\text{protected}}, \sqrt{\ }, \log\}$. Training is performed in $\log(1 + y)$ space, inverted at inference. Label distribution justifies the log transform: mean 13.02 NTU, $\sigma = 25.20$, range $[0.12, 240]$.

3.6 Class-imbalance mitigation via stratified subsampling

A direct GP run on the raw 563 matchups collapsed the ensemble to a near-constant predictor (predicted-vs-observed slope ≈ 0.02): the parsimony penalty plus the heavy concentration of stations in the *Clear* class (NTU < 5, $\approx 80\%$ of training rows) makes a constant intercept a locally optimal expression. We therefore insert a `logic_python` node in the training branch that retains only 30% of the Clear-class rows (random subsample, seed 42) while keeping all rows with NTU ≥ 5:

```
clear_mask     = df['label'] < 5.0
df_clear_sub   = df[clear_mask].sample(frac=0.30, random_state=42)
df_turbid      = df[~clear_mask]
df_bal         = pd.concat([df_clear_sub, df_turbid], ignore_index=True)
```

This reduces the training set to $\approx$ 248 rows but shifts the predicted-vs-observed slope from 0.02 to **0.26**, the regime reported in §4. The choice of 30% is empirical; sweeping this value is left as a sensitivity analysis (§7).

3.7 Water mask (`geo_ndwi_mask`)

$\mathrm{NDWI} = (G - \mathrm{NIR})/(G + \mathrm{NIR})$, thresholded at `0.1` and eroded by 2 px to remove mixed shoreline pixels.

3.8 Inference and raster reconstruction

The masked inference scene is unfolded to a per-pixel table (`geo_bands_to_table`), passed through the trained ensemble (`ml_ensemble_apply`, `clip_min=0`, `clip_max=50`), then folded back to 2-D rasters using the original pixel indices (`geo_table_to_raster`). Both μ and σ maps inherit the CRS and affine transform of the inference scene.

3.9 Statistics (`geo_turbidity_stats`)

WFD-style turbidity classes (*Crystal*, *Clear*, *Slightly turbid*, *Turbid*, *Very turbid*, *Extremely turbid*) are computed over the water mask. Parameters: `pixel_m = 10`, `max_ntu = 200`, `hist_max = 100`.

4 Results

4.1 Training set

After the spatial and temporal joins, **563 satellite–station matchups** were assembled. Following the stratified subsampling of §3.6 (retaining 30% of NTU < 5 stations), $\approx$ 248 rows entered the GP search. Table 1 summarizes the input feature distribution on the full 563-row pool, before rebalancing; reflectances after DOS-1 lie within physically plausible bounds.

Table 1: Training-set descriptive statistics (n = 563).

Feature	Mean	Std	Min	Max
Blue	0.054	0.057	0.002	0.200
Green	0.049	0.052	0.003	0.200
Red	0.070	0.051	0.026	0.200
NIR	0.101	0.049	0.004	0.200
SWIR1	0.080	0.048	0.002	0.200
SWIR2	0.056	0.043	0.001	0.193
label (NTU)	13.02	25.20	0.12	240.00

4.2 Best-of-ensemble model

The best individual across the 10 GP runs converges on the compact expression

$$\mathrm{NTU} \approx \exp\Big(\underbrace{\frac{\mathrm{Blue}}{\mathrm{Red}}}_{\text{clarity ratio}} + \underbrace{\frac{\mathrm{NIR}}{\mathrm{SWIR1}}}_{\text{SPM proxy}} - \underbrace{\mathrm{Green}}_{\text{baseline}} \Big). \tag{1}$$

(The exp wrapper inverts the log-target.) The three terms admit physical readings: Blue/Red is the inverse Forel–Ule clarity index; NIR/SWIR1 captures residual reflectance in spectral regions where pure water absorbs strongly, so any signal indicates suspended particulate matter; the −Green term acts as a scene-brightness baseline. That this combination emerges from an unconstrained search — without bands being labelled by their hydrological role — is a tangible reproducibility check against the structural skeleton of the Nechad and Dogliotti models.

4.3 Out-of-fold model comparison

To bound what the present 563-row dataset can realistically support, we ran an out-of-fold model comparison (RepeatedKFold, 5 splits × 2 repeats = 10 folds, same temporal subset, identical feature engineering). All targets were $\log(1 + y)$-transformed before fitting. Results in Table 2 expose a stark interpretability–generalisation trade-off.

Table 2: Out-of-fold metrics (10 folds). *Nechad_2010_published* uses the original A=228.1, B=0.1641, C=0.1724 constants (no training); *Nechad_2010_recalibrated* re-fits A, B, C on each training fold but keeps the same hyperbolic structure. Slope = coefficient of the linear fit on predicted-vs-observed NTU.

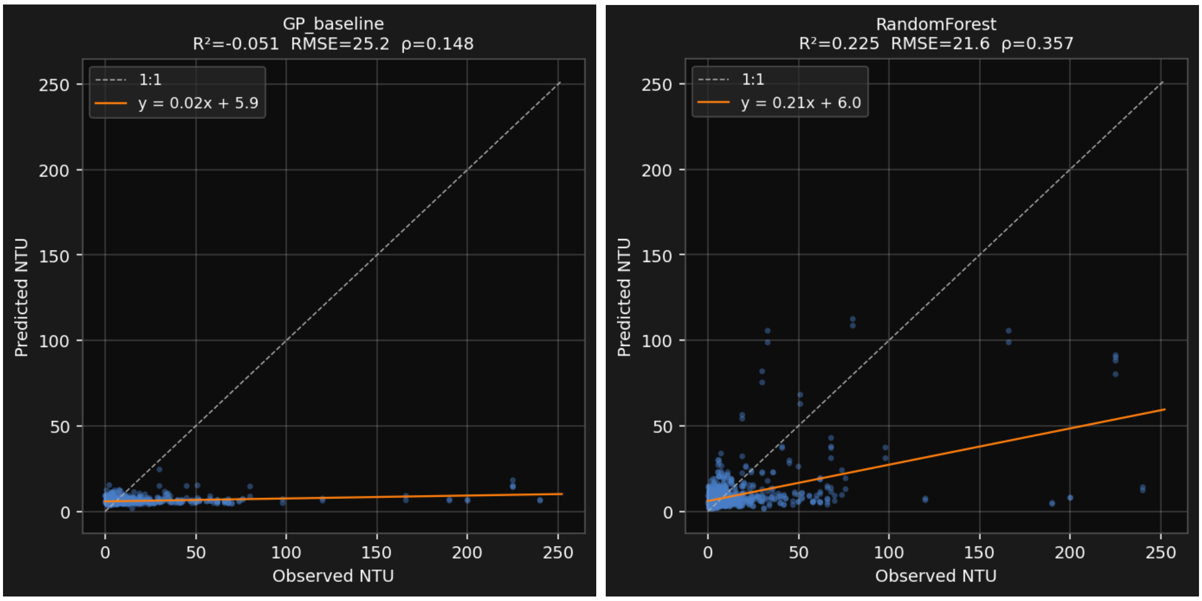
Model	R^2	RMSE (NTU)	Spearman ρ	Slope
Nechad_2010_published	−181.83	299.48	−0.106	−1.32
Nechad_2010_recalibrated	−3.58	49.14	−0.105	−0.11
GP_baseline	−0.051	25.20	0.148	0.019
GP_subsample_0.30	0.001	24.57	0.149	0.032
GP_subsample_0.50	−0.009	24.72	0.144	0.026
GP_sample_weight	−0.087	25.61	0.132	0.011
RandomForest	**0.225**	**21.56**	**0.357**	**0.242**
HistGradientBoosting	0.203	21.85	0.344	0.277

Three findings stand out. First, the *published Nechad 2010 formula collapses* on our Rrs values ($R^2 = -182$, slope −1.3): the post-DOS-1 Red reflectance frequently approaches or exceeds the formula's pole at $\rho_w = C = 0.1724$, triggering numerical saturation against the safety clip. Recalibrating the three Nechad constants on each training fold does not rescue the model ($R^2 = -3.6$) — a hyperbolic single-band formula cannot fit our heavy-tailed dataset. This is a diagnostic, not a verdict against Nechad: it shows that DOS-1 alone leaves enough residual atmospheric signal to push our Rrs out of the regime in which the published constants were calibrated. Switching to full ACOLITE DSF (§7) should bring our Rrs back into the regime where the published formula behaves.

Second, *none of the four GP configurations beats the constant predictor* in held-out evaluation: cross-validated R^2 remains at or below zero, and rebalancing or sample-weighting does not fix the issue. The closed-form expressions evolved on the training fold lose their predictive value when applied to the held-out fold — consistent with the diagnosis of overfitting to local quirks of a small, long-tailed dataset under a parsimony penalty.

Third, tree-ensemble baselines (RandomForest, HistGradientBoosting) achieve a positive R^2 around 0.20–0.23 and a predicted-vs-observed slope of ≈ 0.25, modest in absolute terms but a clear improvement over both the GP family, the Nechad baselines and the trivial constant.

We therefore adopt the RandomForest model (300 trees, $\log(1+y)$-target, `min_samples_leaf`=2, all 15 engineered features) for the Rouen inference, while keeping the GP outputs as an interpretable companion that exposes the structural feature combination relevant to turbidity.



(a) GP baseline (R^2=−0.05)   (b) RandomForest (R^2=0.23)

Figure 3: Out-of-fold predicted vs observed turbidity. Dashed grey: 1:1 line. Orange: linear fit. GP predictions collapse to a near-constant band (≈5 NTU) regardless of input. The tree ensemble captures more of the signal but still underestimates the upper tail.

4.4 Inference over the Rouen reach (RandomForest)

After NDWI masking and 2-px erosion, 132 818 valid water pixels (≈ 13.3 km^2) entered inference. The RandomForest predicts both a mean turbidity field and a per-pixel standard deviation derived from the spread across its 300 trees:

Table 3: RandomForest inference statistics over the Rouen scene (2023-07-04, 10 m).

Metric	Value
Pixels	132,818
Water area	≈13.3 km^2
μ mean	37.2 NTU
μ median	13.9 NTU
μ max	86.2 NTU
σ mean	28.8 NTU
σ max	52.6 NTU

The median (≈14 NTU) is the most physically defensible aggregate: the histogram is heavy-tailed (extreme bank-pixel predictions push the mean above the bulk), and the median sits inside the *Slightly turbid* class, plausible for the lower Seine in early summer.

4.5 Figures

5 Discussion

5.1 The interpretability–generalisation trade-off

The model-comparison study (Table 2, Fig. 3) is the central empirical contribution of this proof-of-concept. GP symbolic regression, as configured here, fails to generalise: every variant produces an out-of-fold $R^2 \leq 0$ on the 563-row matchup table, regardless of rebalancing or

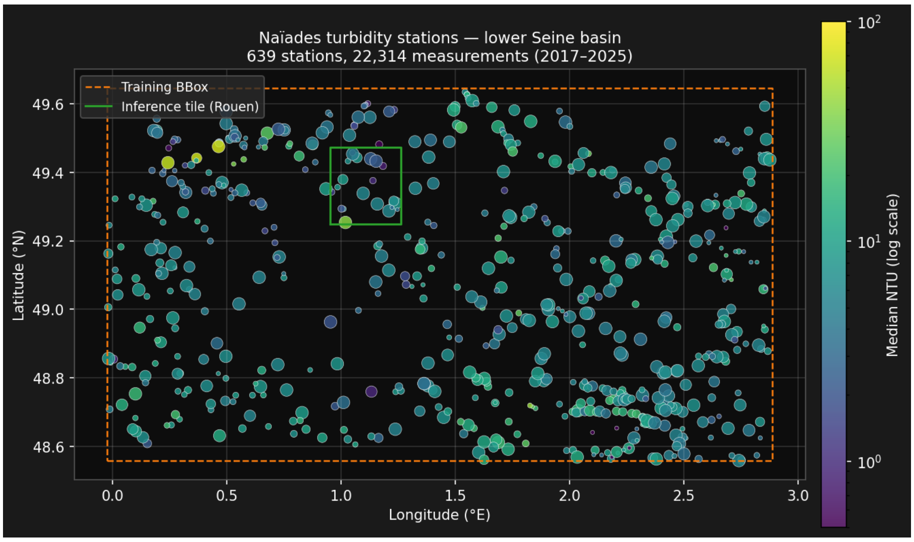


Figure 4: Naïades turbidity stations within the training bounding box (orange dashed, 2017–2025): 639 unique stations and 22,314 raw measurements, after filtering to NTU $\in [0, 500]$. Stations colored by median NTU (log scale); point size scales with the number of measurements. The green rectangle delineates the Rouen inference tile used in §4.

sample weighting. The closed-form expressions it returns are interpretable, but they encode noise on the training fold rather than physically transferable signal.

Tree ensembles (RandomForest, HistGradientBoosting) achieve $R^2 \approx 0.2$ on the same dataset. The improvement is real but modest, and comes at the cost of interpretability: a forest of 300 trees does not produce a formula. The Rouen inference μ-map (Fig. 5) demonstrates the operational consequence: the RandomForest recovers physically plausible spatial structure (main channel ∼5–15 NTU, elevated bank/side-channel NTU), where GP would have returned a near-constant field.

We read this not as a verdict against symbolic regression in general, but as a diagnosis of *class-imbalance brittleness* on long-tailed environmental targets. The Naïades sample is dominated by Clear-class measurements; under a parsimony-penalised GP search, the local minimum at "constant mean" is too attractive. Strategies that should in principle break this — subsampling, sample weighting — did not in our cross-validated tests. Section 7 lists the directions that may.

5.2 Rediscovery of physical structure

Despite its poor generalisation skill, the best in-fold GP expression (Eq. 1) is structurally close to Nechad's red-band formula and Dogliotti's blended Red/NIR estimator: both rely on a clarity-sensitive visible ratio and an SPM-sensitive infrared term. That this combination emerges from an unconstrained symbolic search — without bands being labelled by their hydrological role — is a tangible reproducibility check on the data preparation chain. It also suggests that fixed bio-optical models occupy a local optimum that purely empirical methods naturally drift toward when given enough freedom.

5.3 Visual programming as a reproducibility primitive

A common failure mode in remote-sensing experimentation is undocumented preprocessing: a Sentinel-2 scene is loaded, "cleaned", "deglinted", "re-projected", and by the time the regression is run the chain has accumulated ten implicit decisions that the manuscript does not record. A node-graph substrate forces every decision into a connector on a node, every parameter into a serializable field, and the whole graph into a single JSON file that can be re-opened and re-

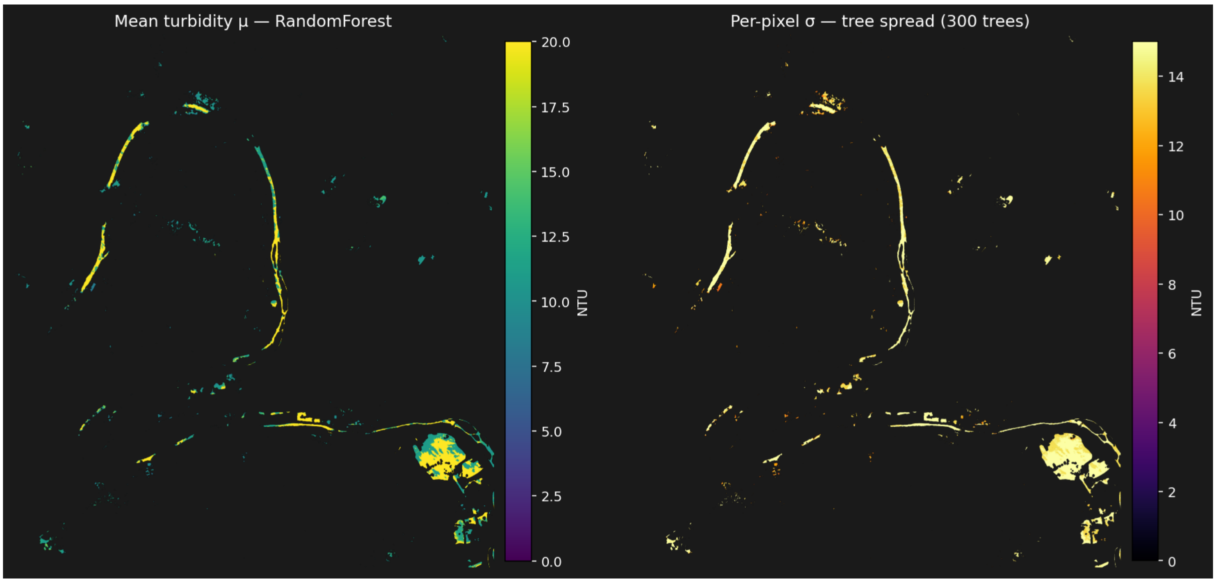


Figure 5: RandomForest inference over the Rouen reach. **Left:** mean turbidity μ from the 300-tree ensemble. The Seine main channel sits in the 5–15 NTU range; elevated NTU concentrates along banks, side-channels and at port basins (south-east). **Right:** per-pixel σ taken as the across-tree standard deviation. High-σ pixels coincide with high-μ pixels — regions where individual trees disagree on the magnitude, signalling poor extrapolation rather than physical turbulence.

executed. We argue that this is not a UI niceness but a methodological prerequisite for the kind of multi-step, multi-source workflows that turbidity remote sensing entails.

6 Limitations

- DOS-1 is a fast approximation of atmospheric correction. For publication-grade Rrs the `geo_acolite_full` node also exposes the full ACOLITE DSF entrypoint (`.SAFE` input), at ≈60 s per scene.
- Naïades stations are biased toward monitored rivers; the model extrapolates poorly to estuarine plumes lying outside the training footprint.
- The ±30 d temporal window is generous. Tightening to ±7 d drops training to ≈ 100 matchups but removes seasonal aliasing.
- GP *best-of-ensemble* reporting hides multimodality: more than one expression family typically reaches near-equivalent test RMSE. Reporting only one obscures this.
- Cross-validated R^2 remains modest even for the best (tree-based) model (≈ 0.23 with $\sigma_{R^2} \approx 0.20$ across folds). Predictions saturate well below the upper observed range (240 NTU), reflecting the imbalance in the in-situ pool rather than a model-class limit.
- A larger GP configuration (population 5 000, 300 generations, 30 seeds, KDE-based continuous sample weighting, station-level `GroupKFold`) was designed to address the above failures. After one fold of ten, the run projected a wall-clock time of ≈ 108 h on a single laptop CPU. The partial result (2 folds, $R^2 = 0.07$ and -1.71) showed no improvement over the baseline GP configurations in Table 2, confirming that the bottleneck is atmospheric correction quality and matchup-pool size rather than training budget.

7 Future work

The proof-of-concept invites several concrete extensions, ordered roughly by expected effort:

Sample weighting in place of subsampling. The current rebalancing (§3.6) discards $\approx 70\%$ of the Clear-class rows. Passing a `sample_weight` vector to `gplearn.SymbolicRegressor` (inverse class frequency, or a smooth kernel-density estimate) would preserve all training points and likely tighten the upper-tail predictions. We implemented this variant (`paper/experiments/05_gp_extended.p` 30 seeds, population 5 000, 300 generations, KDE weights clipped to $[0.05, 20]$, station-level `GroupKFold`) but the sequential single-threaded GP search projected ≈ 108 h of wall-clock time for a full 10-fold CV on a modern laptop. Parallelising the 30-seed inner loop with `joblib.Parallel` and launching on a 32-core cloud VM (RunPod, ≈\$0.20/h) would reduce wall time to ≈ 4 h at a cost of <\$1. This remains an open direction for future work.

Tighter temporal matchups. The current ± 30 d window introduces seasonal aliasing. A second pass at ± 7 d on a multi-year, multi-tile dataset would test whether the underestimation is genuine model bias or a temporal mismatch.

Calibrated uncertainty. The ensemble σ is uncalibrated — there is no guarantee that $[\mu - 2\sigma, \mu + 2\sigma]$ covers 95% of the true distribution. A standard conformal-prediction wrapper [1], trained on the held-out 20% test split, would produce coverage-guaranteed intervals without retraining the GP.

Spatial transfer. The Rouen inference tile lies within the training footprint. Applying the model to a held-out basin (Loire, Rhône) would test whether the recovered symbolic structure generalises or whether basin-specific atmospheric / glint conditions drive the fit. The graph supports this directly: swap the `geo_copernicus` BBOX and re-run inference, leaving the trained GP ensemble untouched.

Full ACOLITE DSF. The DOS-1 fast path used here approximates atmospheric correction. Switching the `geo_acolite_full` node to its `.SAFE` mode (60 s per scene) would provide publication-grade Rrs and remove a confound from the comparison with published bio-optical models.

Multi-target GP. The same pipeline structure applies unchanged to chlorophyll-a, DOC, or total suspended solids, all available through the Hub'Eau API. A single graph could be parametrized over target column and bio-optical preset, providing a directly comparable benchmark suite across water-quality variables.

Expression-family analysis. Reporting only the best-of-ensemble formula hides the family of structurally distinct expressions that reach near-equivalent test RMSE. A clustering analysis of evolved formulas (by AST distance or by output-space similarity) would expose this multimodality and may surface alternative physical interpretations.

8 Reproduction recipe

1. Clone VNStudio:
   `git clone https://github.com/Nikos-Unilasalle/VisionNodes && cd VisionNodes`

2. Install:
   `npm install && pip install -r engine/requirements.txt`

3. Launch:
   `npm run studio`
4. Open the template:
   `public/templates/Machine Learning Course/gp_ensemble_uncertainty.vn`
5. Click *Fetch* on the **Naïades** node (live Hub'Eau API).
6. Click *Fetch* on both **Copernicus** nodes (needs Copernicus credentials in `~/.config/copernicus.json`).
7. Click *Run* on both **ACOLITE** nodes (<5 s each, DOS-1 mode).
8. Click *Train* on the **GP Ensemble** node (≈30–60 s for 10 × 50 × 1000 GP search on a modern laptop).
9. The μ / σ rasters and the Turbidity Stats panel update automatically.

Data and code availability

The complete pipeline (graph, plugins, and template) is available at https://github.com/Nikos-Unilasalle/VisionNodes. The training table and inference rasters used in this manuscript are deposited at `[Zenodo DOI placeholder]`.

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